Sociotechnical Security

CS 8803/4803

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Lecture 19 – Digital intimacy II



Content warning: repeated mentions of sexual abuse, non-consensual intimate imagery, and CSAM

Last time

- Overview of digital intimacy research in sociotechnical security
  - Problems and research questions focused on online dating
  - Problems and research questions focused on sex work
- A generalization across groups: algorithmically-mediated offline introductions
- Discussion
  - How data privacy harms may lead to physical and emotional harms
  - Approaches for giving online daters greater agency over privacy decision-making
  - Consensual vetting?
    - Maybe there are interesting crypto solutions here?

Today

- AI-enabled sexual abuse
  - Terms
  - Research overview
- Proposed solutions and debates
- Overview of policy discussions in the US on AI-enabled sexual abuse
- Ishika will be leading further discussion on the readings

Terminology

- **Image-based sexual abuse (IBSA):** occurs when someone takes, creates, distributes, or threatens to share sexually explicit or intimate photos or videos of a person without their consent
  - NCII
  - NDII
- **AI-enabled sexual abuse**: the use of artificial intelligence technologies to create, distribute, or blackmail individuals with sexually explicit material without their consent
- **Synthetic non-consensual intimate imagery (AIG-NCII):** a form of image-based sexual abuse that includes the non-consensual creation, distribution, and threat of creation or distribution of intimate images.
  - Deepfake nudes
- **AI-generated child sexual abuse material (AIG-CSAM):** **i**mage, video, text, or audio content depicting the sexual abuse or exploitation of minors that has been created either entirely or partially by generative artificial intelligence systems

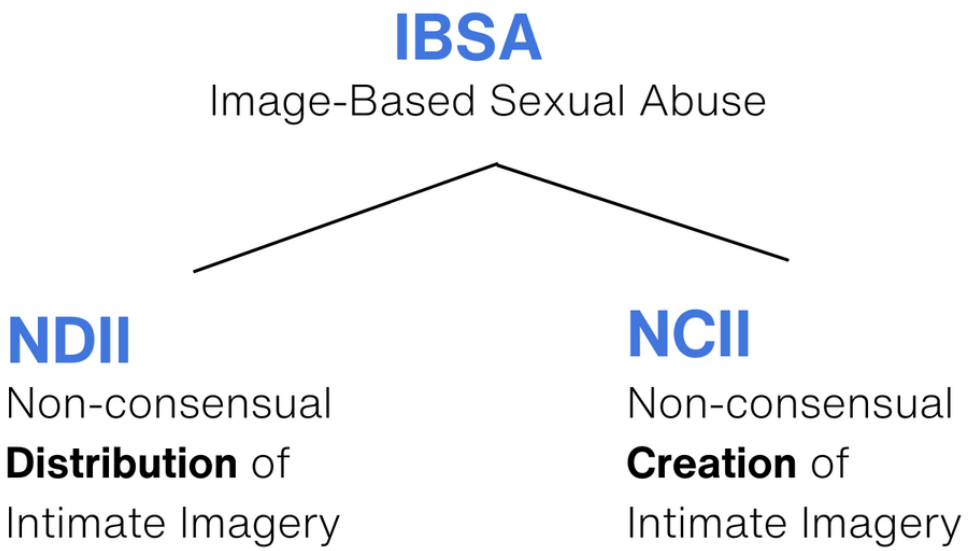


Image credit: Elissa Redmiles

Hard to measure prevalence but some stats:

- ~0.7% of surveyed US Americans have created AIG-NCII
- ~7% have seen AIG-NCII (Umbach et al., CHI 2024)

Viewing AI-NCII is perceived as acceptable by many

- From a survey of 315 US Americans: **~90%** perceive **creating and sharing** of AI-NCII as **unacceptable**; just around **50%** perceive **seeking it out** to view as **unacceptable**. (Brigham et al., SOUPS 2024)

Slide adapted from Elissa Redmiles

Technology as an abuse vector

1

“Nudification” applications

These websites and applications purport to use computer vision techniques to undress the subject of an uploaded image.

2

Generative Models

People may use generative models hosted online to generate AIG-NCII of an individual by e.g., inputing a URL to an image of them as part of their prompt or they may run and fine tune their own local models.

3

Hiring experts: Face swap + custom toolkits

Perpetrators may also hire experts who use a combination of face swapping, generative models, and image editing tools to create AIG-NCII of a target.

Recent research on *videos* discovers 161 GitHub repositories, 60 of which link to academic papers (Han, Li, Kumar & Durumeric 2024).

Slide credit: Elissa Redmiles

Overview of research that folks have done

- Prevalence, attitudes, and norms
- Ecosystem and supply chain
- Perpetrators and creators
- Response
- Youth and schools
- Majority world experiences with AIG-NCII

Proposed solutions

- Model guardrails (stop creation)
  - Training data
  - Input filters
- Deterrence messaging (stop sharing)
  - Intervene when someone is about to search for, share, or upload abusive content
- Proactively identifying NCII (removal)
  - Perceptual hashing
  - Facial recognition web scanning tools

Perceptual hashing

StopNCII.org
Stop Non-Consensual Intimate Image Abuse

1

Victim selects the image

Chosen on their own phone or computer. The image never leaves the device.

2

A hash is made on-device

An algorithm turns the image into a short "digital fingerprint."

3

Only the hash is shared

It goes into a shared hash bank. The victim gets a case number to track or withdraw it.

4

Platforms check for matches

Participating sites hash their content and compare it against the bank, now and on future uploads.

5

Matches are removed or blocked

If the content violates the platform's policy, it comes down and re-uploads are stopped.

Apple CSAM client-side scanning (2021)



AIG-NCII makes this much challenging

- These approaches assume that the victim has the real image that gets copied and shared
- AI allows abusers to create new images that use victims' likeness
  - A victim might not have access to these images

Takedown requests

Request removal of nudity or sexual content from Google Search

We understand that it can be distressing when Google displays search results containing nudity or sexually explicit content related to you.

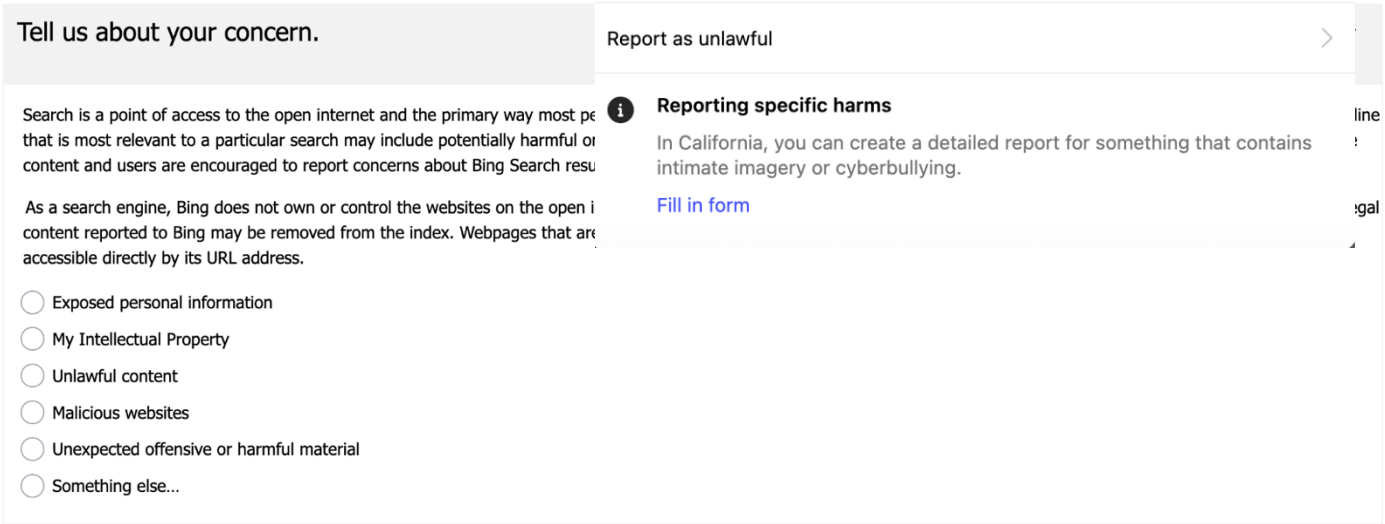
Google may be able to help you or someone you're authorized to represent remove content from Search that:

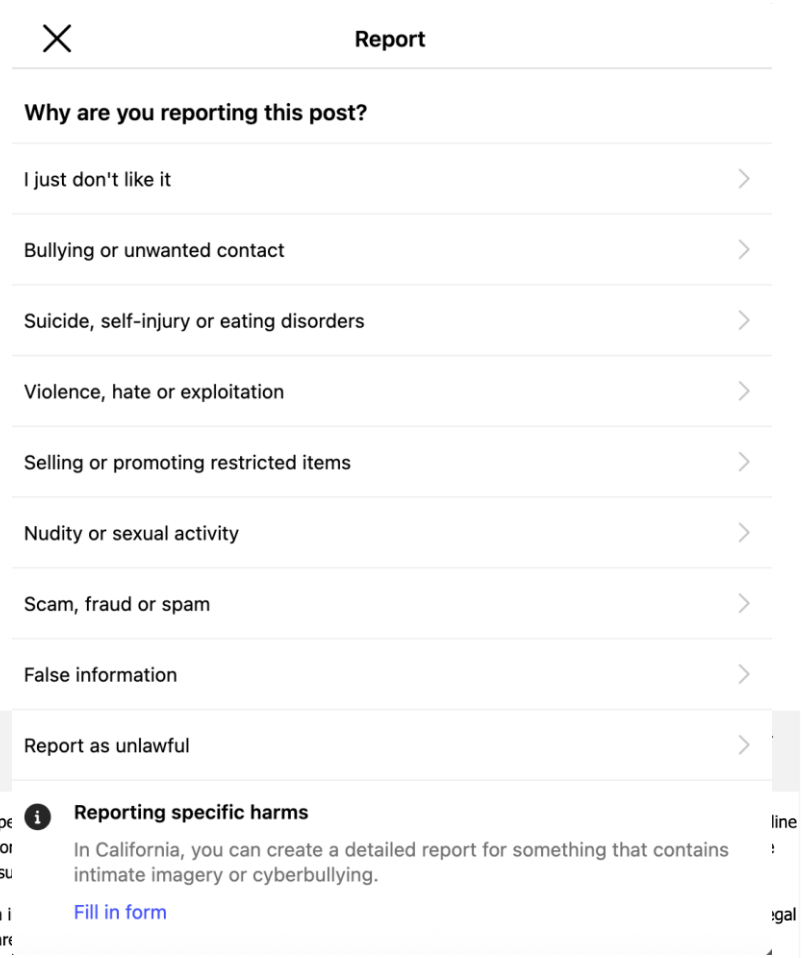
- is nude, intimate, or sexually explicit (for example, images or videos of you) and is distributed without permission; OR
- is fake or falsely depicts you as nude or in a sexually explicit situation; OR
- incorrectly associates you or your name with sex work

To get started, we'll ask you for some information about the content to better understand your situation and get you to the right place to request removal.

Describe the content *

- ◯ Content shows me nude, in a sexual act, or in an intimate state. (This may include, "revenge porn.")
- ◯ Content falsely portrays me in a sexual act, or in an intimate state. (This is sometin fake" or "fake pornography.")
- ◯ Content incorrectly associates me with sexual content
- ◯ Content depicts an individual under the age of 18 in a nude or sexually explicit mar





TAKE IT DOWN Act (Federal)



- Initially signed into law in 2025. It is a federal crime to:
  - Knowingly publish NCII (either real or AIG-NCII)
  - Threaten to publish this content
- First conviction this year
  - Ohio man used AI to create NCII of adults and kids in his neighborhood and shared them on a site promoting CSAM → 15 years in prison in Sep. 2026
- Platform requirements went into effect in May 2026
  - Social media and dating platforms have 48 hours after a valid request to remove NCII
  - Victims can report platforms that have failed to act to the FTC
  - Can also report platforms that do not have a process for requesting removal

MN ban on nudification apps

- First law in the US to target nudification tools
- Prohibits:
  - Operators of websites/apps from letting users nudify images or videos of real people
  - Operators from nudifying content on users' behalf
  - Advertising such applications and services
- Applies to companies, not users
- Applies only to tools that do not require specialized/technical skill to nudify
- Enforcement was paused three days before the law went into effect

Extra credit opportunity: Attend SCP Colloquium on Nov 12 11am

https://sites.gatech.edu/talksatscp/?page_id=8

Speaker: Munmun De Choudhury (http://www.munmund.net/)

Professor of Interactive Computing at GaTech

- Talk is on Nov 12 at 11am in the Coda 9th floor Atrium
- Free lunch at the talk!
- Registration link will be posted at the website soon (and on Ed)
- **If you attend and say hi to Veronica after the talk, you'll get 5 points added to the reading reflection due that week**
- Email me by Monday morning if you don't have Coda access and think you'll come.

